

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1057

National City 138 kV
June 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1057. The Interconnection Customer proposes a 109.0 MW generating facility interconnecting at the National City 138 kV substation in the Virginia area. The request was received on September 02, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 310 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,512,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Decherd to Atoka 138kV Line
Contingency: Spring Crk Road to Atoka 138kV
Loading: 116.5% (Rating: 225 MVA)
DFAX: 70.6%
Pre-project loading: 82.3%
2. Facility: Coyote to Hobart 138kV L
Contingency: Carlton Hills to Clinch River 345kV
Loading: 110.8% (Rating: 253 MVA)
DFAX: 7.2%
Pre-project loading: 107.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Atoka to Decherd 138kV Line	Rebuild line with higher-capacity conductor	\$4,678,000	40% (\$1,874,000)
Coyote to Hobart 138kV L	Rebuild line with higher-capacity conductor	\$5,137,000	32% (\$1,638,000)
	Total allocated to project		\$3,512,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.